Concept Question 5-4: If one of two wires of equal length is formed into a closed square loop and the other into a closed circular loop, and if both wires are carrying equal currents and both loops have their planes parallel to a uniform magnetic field, which loop would experience the greater torque?

Exercise 5-5: A square coil of 100 turns and 0.5-m long sides is in a region with a uniform magnetic flux density of 0.2 T. If the maximum magnetic torque exerted on the coil is 4×10^{-2} (N·m), what is the current flowing in the coil?

Answer: $I = 8$ mA. (See EM.)

5-2 The Biot–Savart Law

In the preceding section, we elected to use the magnetic flux density **B** to denote the presence of a magnetic field in a given region of space. We now work with the magnetic field intensity **H** instead. We do this in part to remind the reader that for most materials the flux and field are linearly related by $\mathbf{B} = \mu\mathbf{H}$; therefore, knowledge of one implies knowledge of the other (assuming that μ is known).

Through his experiments on the deflection of compass needles by current-carrying wires, Hans Oersted established that currents induce magnetic fields that form closed loops around the wires (see Section 1-3.3). Building upon Oersted's results, Jean Biot and Félix Savart arrived at an expression that relates the magnetic field **H** at any point in space to the current I that generates **H**. The ***Biot–Savart law*** states that the differential magnetic field $d\mathbf{H}$ generated by a steady current I flowing through a differential length vector $d\mathbf{l}$ is

$$d\mathbf{H} = \frac{I}{4\pi}\frac{d\mathbf{l} \times \hat{\mathbf{R}}}{R^2} \quad \text{(A/m)}, \tag{5.21}$$

where $\mathbf{R} = \hat{\mathbf{R}}R$ is the distance vector between $d\mathbf{l}$ and the observation point P, as shown in **Fig. 5-8**. The SI unit for **H** is ampere·m/m^2 = (A/m). It is important to remember that Eq. (5.21) assumes that $d\mathbf{l}$ is along the direction of the current I and the unit vector $\hat{\mathbf{R}}$ points *from* the current element *to* the observation point.

According to Eq. (5.21), $d\mathbf{H}$ varies as R^{-2}, which is similar to the distance dependence of the electric field induced by an electric charge. However, unlike the electric field vector **E**, whose direction is *along* the distance vector **R** joining the charge to the observation point, the magnetic field **H** is *orthogonal* to the plane containing the direction of the current element $d\mathbf{l}$ and the distance vector **R**. At point P in **Fig. 5-8**, the direction of $d\mathbf{H}$ is out of the page, whereas at point P' the direction of $d\mathbf{H}$ is into the page.

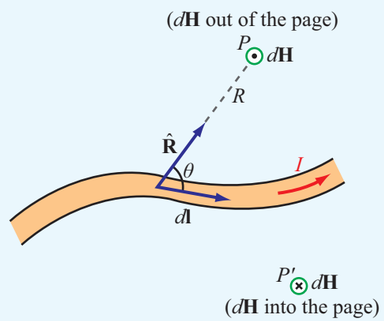


Figure 5-8 Magnetic field $d\mathbf{H}$ generated by a current element $I\,d\mathbf{l}$. The direction of the field induced at point P is opposite to that induced at point P'.

To determine the total magnetic field **H** due to a conductor of finite size, we need to sum up the contributions due to all the current elements making up the conductor. Hence, the Biot–Savart law becomes

$$\mathbf{H} = \frac{I}{4\pi}\int_l \frac{d\mathbf{l} \times \hat{\mathbf{R}}}{R^2} \quad \text{(A/m)}, \tag{5.22}$$

where l is the line path along which I exists.

5-2.1 Magnetic Field Due to Surface and Volume Current Distributions

The Biot–Savart law also may be expressed in terms of distributed current sources (**Fig. 5-9**) such as the ***volume current density*** **J**, measured in (A/m^2), or the ***surface current density*** $\mathbf{J}_s$, measured in (A/m). The surface current density $\mathbf{J}_s$ applies to currents that flow on the surface of a conductor in the form of a sheet of effectively zero thickness. When current sources are specified in terms of $\mathbf{J}_s$ over a surface S or in terms of **J** over a volume $\mathcal{V}$, we can use the equivalence given by

$$I\,d\mathbf{l} \leftrightarrow \mathbf{J}_s\,ds \leftrightarrow \mathbf{J}\,d\mathcal{V} \tag{5.23}$$

概念题 5-4：将两根等长导线分别弯成闭合正方形回路和闭合圆形回路。若两回路中电流相等，且回路平面均平行于匀强磁场，哪个回路受到的力矩更大？

练习 5-5：一个 100 匝、边长为 0.5 m 的正方形线圈置于磁通密度为 0.2 T 的匀强磁场中。若线圈所受最大磁力矩为 4×10^{-2} N·m，求线圈中的电流。

答案：I = 8 mA。（参见配套 EM 资源。）

5-2 毕奥–萨伐尔定律

上一节用磁通密度 B 描述空间中的磁场。现在改用磁场强度 H，以提醒读者：对于大多数材料，二者具有线性关系 B = μH。因此，在磁导率 μ 已知的条件下，知道其中一个量便可确定另一个量。

奥斯特通过载流导线使磁针偏转的实验，发现电流产生的磁场围绕导线形成闭合磁力线（见第 1-3.3 节）。在此基础上，毕奥和萨伐尔得出了空间任意点的磁场 H 与产生该磁场的电流 I 之间的关系。毕奥–萨伐尔定律指出：稳恒电流 I 流经微小有向线元 dl 时，产生的磁场微元 dH 为

$$d\mathbf{H} = \frac{I}{4\pi}\frac{d\mathbf{l} \times \hat{\mathbf{R}}}{R^2} \quad \text{(A/m)}, \tag{5.21}$$

其中，矢量 R 从线元 dl 指向观察点 P，如图 5-8 所示；其长度为 R。H 的 SI 单位为 A/m。式（5.21）中，dl 沿电流 I 的方向，R 上加帽号表示从电流元指向观察点的单位矢量。

由式（5.21）可知，dH 与 R^2 成反比，其距离关系与点电荷产生的电场相似。但电场 E 沿电荷指向观察点的位移矢量 R，而电流元产生的磁场 dH 垂直于 dl 与 R 所在的平面。在图 5-8 中的 P 点，

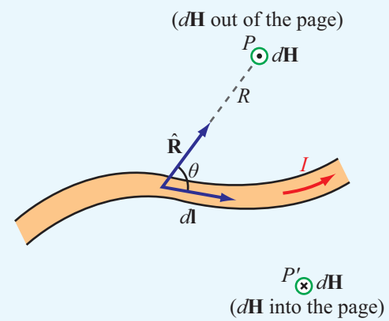


图 5-8 电流元 I dl 产生的磁场 dH。P 点与 P′ 点的磁场方向相反。图中 out of the page 表示垂直纸面向外，into the page 表示垂直纸面向内。

dH 垂直纸面向外；在 P′ 点，dH 垂直纸面向内。

为求有限尺寸导体产生的总磁场 H，需要将组成导体的所有电流元产生的磁场叠加。因此，毕奥–萨伐尔定律可写为

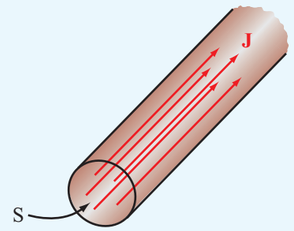
$$\mathbf{H} = \frac{I}{4\pi}\int_l \frac{d\mathbf{l} \times \hat{\mathbf{R}}}{R^2} \quad \text{(A/m)}, \tag{5.22}$$

其中，l 为电流 I 所沿的积分路径。

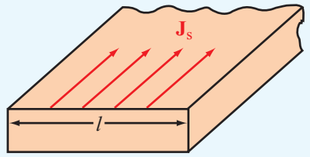
5-2.1 面电流和体电流分布产生的磁场

毕奥–萨伐尔定律也可用分布电流源表示（图 5-9），例如体电流密度 J（单位 A/m^2）或面电流密度 J_s（单位 A/m）。J_s 用于描述沿导体表面流动、可视为厚度为零的薄层电流。当电流源由曲面 S 上的 J_s 或体积 v 内的 J 给定时，可采用以下等效关系：

$$I\,d\mathbf{l} \leftrightarrow \mathbf{J}_s\,ds \leftrightarrow \mathbf{J}\,d\mathcal{V} \tag{5.23}$$



(a) Volume current density **J** in A/m^2



(b) Surface current density $\mathbf{J}_s$ in A/m

Figure 5-9 (a) The total current crossing the cross section S of the cylinder is $I = \int_S \mathbf{J} \cdot d\mathbf{s}$. (b) The total current flowing across the surface of the conductor is $I = \int_l J_s \, dl$.

to express the Biot–Savart law as

$$\mathbf{H} = \frac{1}{4\pi} \int_S \frac{\mathbf{J}_s \times \hat{\mathbf{R}}}{R^2} \, ds, \tag{5.24a}$$

(surface current)

$$\mathbf{H} = \frac{1}{4\pi} \int_v \frac{\mathbf{J} \times \hat{\mathbf{R}}}{R^2} \, dv. \tag{5.24b}$$

(volume current)

Example 5-2: Magnetic Field of a Linear Conductor

A free-standing linear conductor of length l carries a current I along the z axis as shown in **Fig. 5-10**. Determine the magnetic flux density **B** at a point P located at a distance r in the x–y plane. The wire is, of course, part of a closed-loop circuit, but our current interest is only in the conducting wire.

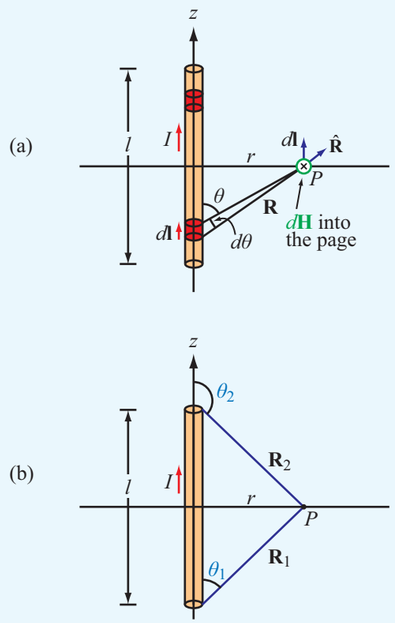


Figure 5-10 Linear conductor of length l carrying a current I. (a) The field $d\mathbf{H}$ at point P due to incremental current element $d\mathbf{l}$. (b) Limiting angles θ_1 and θ_2, each measured between vector $I\,d\mathbf{l}$ and the vector connecting the end of the conductor associated with that angle to point P (Example 5-2).

Solution: From **Fig. 5-10**, the direction of I is along $+\hat{\mathbf{z}}$. Hence, the differential length vector is $d\mathbf{l} = \hat{\mathbf{z}}\, dz$, and $d\mathbf{l} \times \hat{\mathbf{R}} = dz\,(\hat{\mathbf{z}} \times \hat{\mathbf{R}}) = \hat{\boldsymbol{\phi}} \sin\theta \, dz$, where $\hat{\boldsymbol{\phi}}$ is the azimuth direction and θ is the angle between $d\mathbf{l}$ and $\hat{\mathbf{R}}$. Application of Eq. (5.22) gives

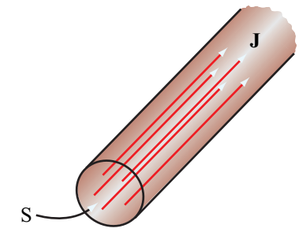
$$\mathbf{H} = \frac{I}{4\pi} \int_{z=-l/2}^{z=l/2} \frac{d\mathbf{l} \times \hat{\mathbf{R}}}{R^2} = \hat{\boldsymbol{\phi}} \frac{I}{4\pi} \int_{-l/2}^{l/2} \frac{\sin\theta}{R^2} \, dz. \tag{5.25}$$

Both R and θ are dependent on the integration variable z, but the radial distance r is not. For convenience, we convert the integration variable from z to θ by using the transformations

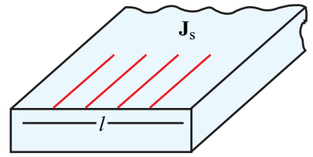
$$R = r \csc\theta, \tag{5.26a}$$

$$z = -r \cot\theta, \tag{5.26b}$$

$$dz = r \csc^2\theta \, d\theta. \tag{5.26c}$$



(a) Volume current density **J** in A/m^2



(b) Surface current density $\mathbf{J}_s$ in A/m

图5-9 (a)穿过圆柱体横截面S的总电流为I= $\int_S$J·ds。 (b) 流过导体表面的总电流为 I = $\int_l$ J_s dl。

将毕奥-萨伐尔定律表示为

$$\mathbf{H} = \frac{1}{4\pi} \int_S \frac{\mathbf{J}_s \times \hat{\mathbf{R}}}{R^2} \, ds, \tag{5.24a}$$

(表面电流)

$$\mathbf{H} = \frac{1}{4\pi} \int_v \frac{\mathbf{J} \times \hat{\mathbf{R}}}{R^2} \, dv. \tag{5.24b}$$

(音量电流)

示例 5-2：线性导体的磁场

长度为 l 的独立线性导体沿 z 轴承载电流 I，如图 5-10 所示。确定 x-y 平面中距离 r 处的点 P 处的磁通密度 B。当然，导线是闭环电路的一部分，但我们当前的兴趣仅在于导线。

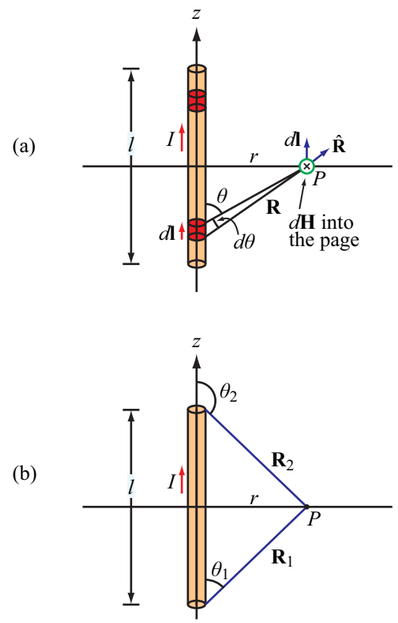


图 5-10 长度为 l 的线性导体，承载电流 I。 (a) 由于增量电流元件 dl，P 点处的磁场 dH。 (b) 限制角度 θ_1 和 θ_2，每个角度在矢量 I dl 和连接与该角度相关的导体末端到点 P 的矢量之间测量（示例 5-2）。

解：由图5-10可知，I的方向是沿+ẑ。因此，差分长度向量为 dl = ẑ dz，dl × R̂ = dz (ẑ × R̂) = $\hat{\boldsymbol{\phi}}$ sin$\hat{\boldsymbol{\theta}}$ dz，其中$\hat{\boldsymbol{\phi}}$是方位角方向，θ 是 dl 和 R̂ 之间的角度。方程的应用(5.22) 给出

$$\mathbf{H} = \frac{I}{4\pi} \int_{z=-l/2}^{z=l/2} \frac{d\mathbf{l} \times \hat{\mathbf{R}}}{R^2} = \hat{\boldsymbol{\phi}} \frac{I}{4\pi} \int_{-l/2}^{l/2} \frac{\sin\theta}{R^2} \, dz. \tag{5.25}$$

R 和 θ 都取决于积分变量 z，但径向距离 r 则不然。为了方便起见，我们使用转换将积分变量从 z 转换为 θ

$$R = r \csc\theta, \tag{5.26a}$$

$$z = -r \cot\theta, \tag{5.26b}$$

$$dz = r \csc^2\theta \, d\theta. \tag{5.26c}$$

Upon inserting Eqs. (5.26a) and (5.26c) into Eq. (5.25), we have

$$\mathbf{H} = \hat{\boldsymbol{\phi}} \frac{I}{4\pi} \int_{\theta_1}^{\theta_2} \frac{\sin\theta \, r\csc^2\theta \, d\theta}{r^2 \csc^2\theta}$$

$$= \hat{\boldsymbol{\phi}} \frac{I}{4\pi r} \int_{\theta_1}^{\theta_2} \sin\theta \, d\theta = \hat{\boldsymbol{\phi}} \frac{I}{4\pi r} (\cos\theta_1 - \cos\theta_2), \quad (5.27)$$

where θ_1 and θ_2 are the limiting angles at $z = -l/2$ and $z = l/2$, respectively. From the right triangle in **Fig. 5-10(b)**, it follows that

$$\cos\theta_1 = \frac{l/2}{\sqrt{r^2 + (l/2)^2}}, \quad (5.28a)$$

$$\cos\theta_2 = -\cos\theta_1 = \frac{-l/2}{\sqrt{r^2 + (l/2)^2}}. \quad (5.28b)$$

Hence,

$$\mathbf{B} = \mu_0 \mathbf{H} = \hat{\boldsymbol{\phi}} \frac{\mu_0 I l}{2\pi r \sqrt{4r^2 + l^2}} \quad \text{(T)}. \quad (5.29)$$

For an infinitely long wire with $l \gg r$, Eq. (5.29) reduces to

$$\mathbf{B} = \hat{\boldsymbol{\phi}} \frac{\mu_0 I}{2\pi r}. \quad \textbf{(infinitely long wire)} \quad (5.30)$$

► This is a very important and useful expression to keep in mind. It states that in the neighborhood of a linear conductor carrying a current I, the induced magnetic field forms concentric circles around the wire (**Fig. 5-11**), and its intensity is directly proportional to I and inversely proportional to the distance r. ◄

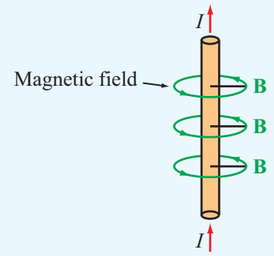


Figure 5-11 Magnetic field surrounding a long, linear current-carrying conductor.

Example 5-3: Magnetic Field of a Circular Loop

A circular loop of radius a carries a steady current I. Determine the magnetic field $\mathbf{H}$ at a point on the axis of the loop.

Solution: Let us place the loop in the x–y plane (**Fig. 5-12**). Our task is to obtain an expression for $\mathbf{H}$ at point $P(0, 0, z)$.

We start by noting that any element $d\mathbf{l}$ on the circular loop is perpendicular to the distance vector $\mathbf{R}$ and that all elements around the loop are at the same distance R from P with $R = \sqrt{a^2 + z^2}$. From Eq. (5.21), the magnitude of $d\mathbf{H}$ due to current element $d\mathbf{l}$ is

$$dH = \frac{I}{4\pi R^2} |d\mathbf{l} \times \hat{\mathbf{R}}| = \frac{I \, dl}{4\pi(a^2 + z^2)}, \quad (5.31)$$

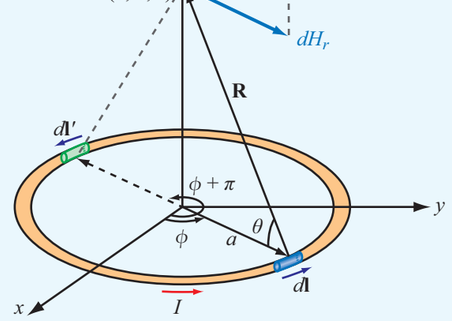


Figure 5-12 Circular loop carrying a current I (Example 5-3).

and the direction of $d\mathbf{H}$ is perpendicular to the plane containing $\mathbf{R}$ and $d\mathbf{l}$. $d\mathbf{H}$ is in the r–z plane (**Fig. 5-12**); therefore, it has components dH_r and dH_z. If we consider element $d\mathbf{l}'$ located diametrically opposite to $d\mathbf{l}$, we observe that the z components of the magnetic fields due to $d\mathbf{l}$ and $d\mathbf{l}'$ add because they are in the same direction, but their r components cancel because they are in opposite directions. Hence, the non-vanishing

插入方程式后。（5.26a）和（5.26c）代入方程。（5.25），我们有

$$\mathbf{H} = \hat{\boldsymbol{\phi}} \frac{I}{4\pi} \int_{\theta_1}^{\theta_2} \frac{\sin\theta \, r\csc^2\theta \, d\theta}{r^2 \csc^2\theta}$$

$$= \hat{\boldsymbol{\phi}} \frac{I}{4\pi r} \int_{\theta_1}^{\theta_2} \sin\theta \, d\theta = \hat{\boldsymbol{\phi}} \frac{I}{4\pi r} (\cos\theta_1 - \cos\theta_2), \quad (5.27)$$

其中 θ_1 和 θ_2 分别是 $z = -l/2$ 和 $z = l/2$ 处的极限角。由图5-10(b)中的直角三角形可知

$$\cos\theta_1 = \frac{l/2}{\sqrt{r^2 + (l/2)^2}}, \quad (5.28a)$$

$$\cos\theta_2 = -\cos\theta_1 = \frac{-l/2}{\sqrt{r^2 + (l/2)^2}}. \quad (5.28b)$$

因此，

$$\mathbf{B} = \mu_0 \mathbf{H} = \hat{\boldsymbol{\phi}} \frac{\mu_0 I l}{2\pi r \sqrt{4r^2 + l^2}} \quad \text{(T)}. \quad (5.29)$$

对于具有 $l \gg r$ 的无限长导线，方程：(5.29) 减少为

$$\mathbf{B} = \hat{\boldsymbol{\phi}} \frac{\mu_0 I}{2\pi r}. \quad \textbf{(infinitely long wire)} \quad (5.30)$$

► 这是一个非常重要且有用的表达方式，需要牢记在心。它指出，在承载电流 I 的线性导体附近，感应磁场在导线周围形成同心圆（图 5-11），其强度与 I 成正比，与距离 r 成反比。◄

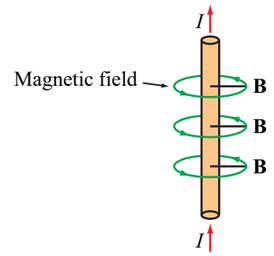


图 5-11 长线性载流导体周围的磁场。

例 5-3：圆环的磁场

半径为 a 的圆环流过稳定电流 I。确定环轴上一点的磁场 H。

解决方案：让我们将环放置在 x-y 平面上（图 5-12）。我们的任务是获得点 P(0, 0, z) 处的 H 表达式。

我们首先注意到圆形环上的任何元素 dl 都垂直于距离向量 R，并且环周围的所有元素与 P 的距离 R 相同，其中 $R = \sqrt{a^2 + z^2}$。从方程。(5.21) 电流元素 dl 引起的 dH 的大小为

$$dH = \frac{I}{4\pi R^2} |d\mathbf{l} \times \hat{\mathbf{R}}| = \frac{I \, dl}{4\pi(a^2 + z^2)}, \quad (5.31)$$

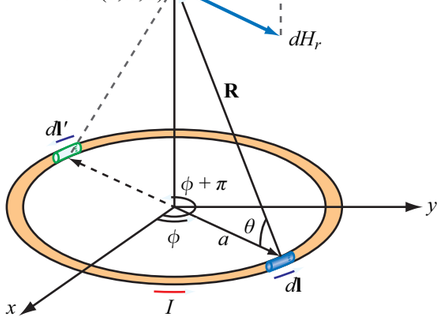
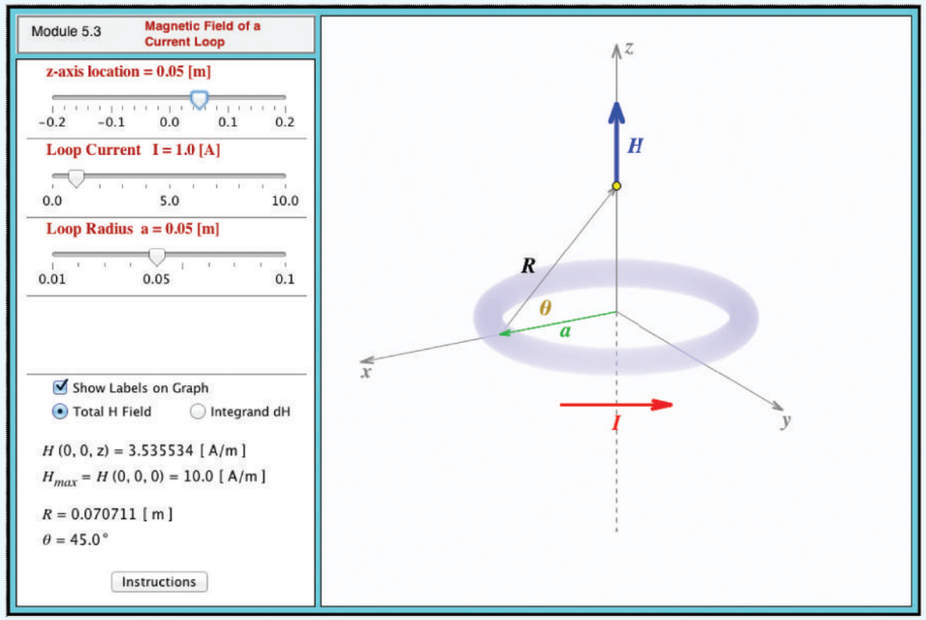


图 5-12 承载电流 I 的环形回路（示例 5-3）。

dH的方向垂直于包含R和dl的平面。 dH 位于 r-z 平面（图 5-12）；因此，它具有分量 dH_r 和 dH_z。如果我们考虑元素 dl′ 与 dl 完全相反，我们会观察到 dl 和 dl′ 产生的磁场 z 分量相加，因为它们方向相同，但它们的 r 分量抵消，因为它们方向相反。因此，不消失的

Module 5.3 Magnetic Field of a Current Loop Examine how the field along the loop axis changes with loop parameters.



component of the magnetic field is along z only. That is,

$$d\mathbf{H} = \hat{\mathbf{z}}\, dH_z = \hat{\mathbf{z}}\, dH \cos\theta = \hat{\mathbf{z}}\, \frac{I\cos\theta}{4\pi(a^2+z^2)}\, dl. \tag{5.32}$$

For a fixed point $P(0,0,z)$ on the axis of the loop, all quantities in Eq. (5.32) are constant, except for dl. Hence, integrating Eq. (5.32) over a circle of radius a gives

$$\mathbf{H} = \hat{\mathbf{z}}\, \frac{I\cos\theta}{4\pi(a^2+z^2)} \oint dl = \hat{\mathbf{z}}\, \frac{I\cos\theta}{4\pi(a^2+z^2)}\, (2\pi a). \tag{5.33}$$

Upon using the relation $\cos\theta = a/(a^2+z^2)^{1/2}$, we obtain

$$\mathbf{H} = \hat{\mathbf{z}}\, \frac{Ia^2}{2(a^2+z^2)^{3/2}} \quad \text{(A/m)}. \tag{5.34}$$

At the center of the loop ($z = 0$), Eq. (5.34) reduces to

$$\mathbf{H} = \hat{\mathbf{z}}\, \frac{I}{2a} \quad (\text{at } z = 0), \tag{5.35}$$

and at points very far away from the loop such that $z^2 \gg a^2$, Eq. (5.34) simplifies to

$$\mathbf{H} = \hat{\mathbf{z}}\, \frac{Ia^2}{2|z|^3} \quad (\text{at } |z| \gg a). \tag{5.36}$$

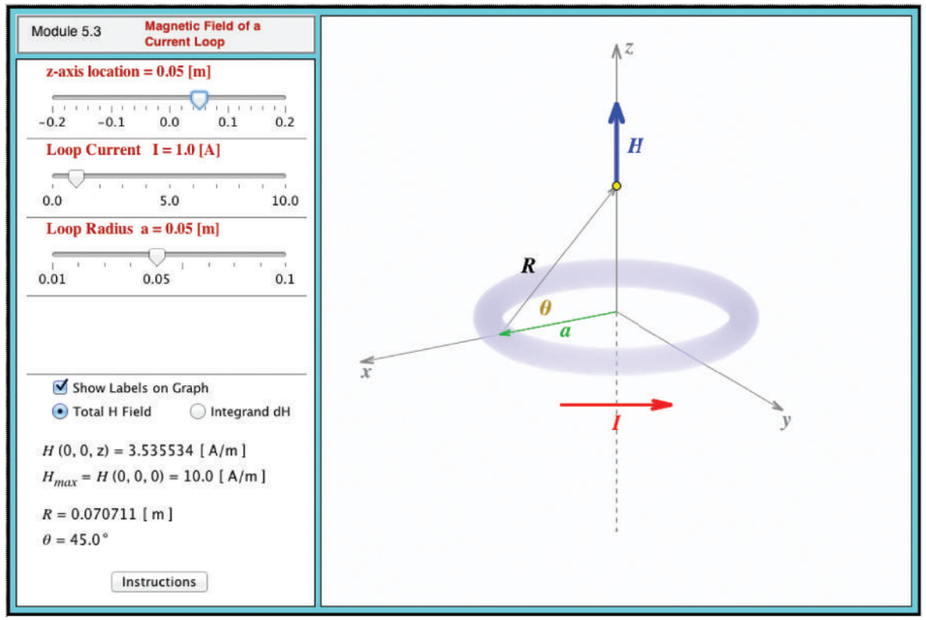
5-2.2 Magnetic Field of a Magnetic Dipole

In view of the definition given by Eq. (5.19) for the magnetic moment $\mathbf{m}$ of a current loop, a single-turn loop situated in the x–y plane (**Fig. 5-12**) has magnetic moment $\mathbf{m} = \hat{\mathbf{z}}m$ with $m = I\pi a^2$. Consequently, Eq. (5.36) may be expressed as

$$\mathbf{H} = \hat{\mathbf{z}}\, \frac{m}{2\pi|z|^3} \quad (\text{at } |z| \gg a). \tag{5.37}$$

This expression applies to a point P far away from the loop and on its axis. Had we solved for $\mathbf{H}$ at any distant point

模块 5.3 电流环路的磁场 检查沿环路轴的磁场如何随环路参数变化。



磁场的分量仅沿 z 方向。那是，

$$d\mathbf{H} = \hat{\mathbf{z}}\, dH_z = \hat{\mathbf{z}}\, dH \cos\theta = \hat{\mathbf{z}}\, \frac{I\cos\theta}{4\pi(a^2+z^2)}\, dl. \tag{5.32}$$

对于循环轴上的固定点 P(0, 0, z)，方程式中的所有量。(5.32) 是常数，除了 dl 之外。因此，整合方程。 (5.32) 在半径为 a 的圆上给出

$$\mathbf{H} = \hat{\mathbf{z}}\, \frac{I\cos\theta}{4\pi(a^2+z^2)} \oint dl = \hat{\mathbf{z}}\, \frac{I\cos\theta}{4\pi(a^2+z^2)}\, (2\pi a). \tag{5.33}$$

使用关系 $\cos\theta = a/(a^2+z^2)^{1/2}$，我们得到

$$\mathbf{H} = \hat{\mathbf{z}}\, \frac{Ia^2}{2(a^2+z^2)^{3/2}} \quad \text{(A/m)}. \tag{5.34}$$

在循环的中心 (z = 0)，方程： (5.34) 减少为

$$\mathbf{H} = \hat{\mathbf{z}}\, \frac{I}{2a} \quad (\text{at } z = 0), \tag{5.35}$$

并且在距离循环很远的点处，使得 $z^2 \gg a^2$，方程： (5.34) 简化为

$$\mathbf{H} = \hat{\mathbf{z}}\, \frac{Ia^2}{2|z|^3} \quad (\text{at } |z| \gg a). \tag{5.36}$$

5-2.2 磁偶极子的磁场

鉴于方程给出的定义。 (5.19) 对于电流环路的磁矩 m，位于 x-y 平面（图 5-12）的单匝环路的磁矩为 $\mathbf{m} = \hat{\mathbf{z}}m$，其中 $m = I\pi a^2$。因此，等式。 (5.36) 可表示为

$$\mathbf{H} = \hat{\mathbf{z}}\, \frac{m}{2\pi|z|^3} \quad (\text{at } |z| \gg a). \tag{5.37}$$

该表达式适用于远离环且位于其轴上的点 P。我们是否可以求解任意远点的 H

$P = (R, \theta, \phi)$ in a spherical coordinate system with R the distance between the center of the loop and point P, we would have obtained the expression

$$\mathbf{H} = \frac{m}{4\pi R^3}(\hat{\mathbf{R}}\,2\cos\theta + \hat{\boldsymbol{\theta}}\sin\theta) \qquad (\text{for } R \gg a). \tag{5.38}$$

▶ A current loop with dimensions much smaller than the distance between the loop and the observation point is called a ***magnetic dipole***. This is because the pattern of its magnetic field lines is similar to that of a permanent magnet as well as to the pattern of the electric field lines of the electric dipole (**Fig. 5-13**). ◀

Concept Question 5-5: Two infinitely long parallel wires carry currents of equal magnitude. What is the resultant magnetic field due to the two wires at a point midway between the wires—compared with the magnetic field due to one of them alone—if the currents are (a) in the same direction and (b) in opposite directions?

Concept Question 5-6: Devise a right-hand rule for the direction of the magnetic field due to a linear current-carrying conductor.

Concept Question 5-7: What is a magnetic dipole? Describe its magnetic field distribution.

Exercise 5-6: A semi-infinite linear conductor extends between $z = 0$ and $z = \infty$ along the z axis. If the current I in the conductor flows along the positive z direction, find $\mathbf{H}$ at a point in the x–y plane at a radial distance r from the conductor.

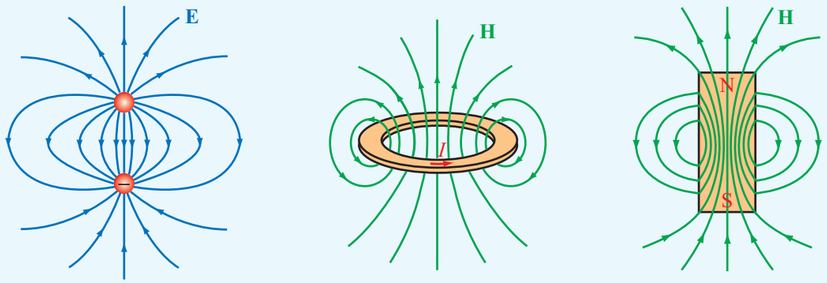
Answer: $\mathbf{H} = \hat{\boldsymbol{\phi}}\dfrac{I}{4\pi r}$ (A/m). (See EM.)

Exercise 5-7: A wire carrying a current of 4 A is formed into a circular loop. If the magnetic field at the center of the loop is 20 A/m, what is the radius of the loop if the loop has (a) only one turn and (b) 10 turns?

Answer: (a) $a = 10$ cm, (b) $a = 1$ m. (See EM.)

Exercise 5-8: A wire is formed into a square loop and placed in the x–y plane with its center at the origin and each of its sides parallel to either the x or y axes. Each side is 40 cm in length, and the wire carries a current of 5 A whose direction is clockwise when the loop is viewed from above. Calculate the magnetic field at the center of the loop.

Answer: $\mathbf{H} = -\hat{\mathbf{z}}\dfrac{4I}{\sqrt{2\pi l}} = -\hat{\mathbf{z}}11.25$ A/m. (See EM.)



(a) Electric dipole (b) Magnetic dipole (c) Bar magnet

Figure 5-13 Patterns of (a) the electric field of an electric dipole, (b) the magnetic field of a magnetic dipole, and (c) the magnetic field of a bar magnet. Far away from the sources, the field patterns are similar in all three cases.

P = (R, θ, ϕ) 在球坐标系中，R 为环中心与点 P 之间的距离，我们可以得到表达式

$$\mathbf{H} = \frac{m}{4\pi R^3}(\hat{\mathbf{R}}\,2\cos\theta + \hat{\boldsymbol{\theta}}\sin\theta) \qquad (\text{for } R \gg a). \tag{5.38}$$

▶ 尺寸远小于环路与观察点之间的距离的电流环路称为磁偶极子。这是因为它的磁场线图案与永磁体的磁场线图案以及电偶极子的电场线图案相似（图5-13）。◀

概念问题 5-5：两条无限长的平行导线承载大小相等的电流。如果电流 (a) 方向相同且 (b) 方向相反，那么与单独一根导线产生的磁场相比，两根导线在导线中间点产生的磁场是多少？

概念问题 5-6：为线性载流导体产生的磁场方向设计右手定则。

概念问题 5-7：什么是磁偶极子？描述其磁场分布。

练习 5-6：半无限线性导体沿 z 轴在 z = 0 和 z = ∞ 之间延伸。如果导体中的电流 I 沿 z 轴正方向流动，则在 x-y 平面上距导体径向距离 r 处的点处找到 H。

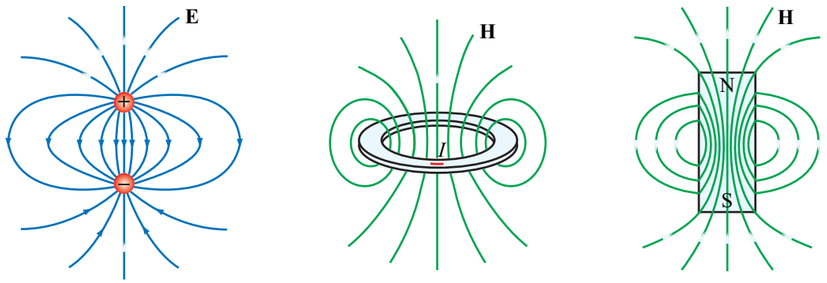
Answer: $\mathbf{H} = \hat{\boldsymbol{\phi}}\dfrac{I}{4\pi r}$ (A/m). (See EM.)

练习 5-7：将承载 4 A 电流的导线形成一个环形回路。如果环路中心的磁场为 20 A/m，则环路 (a) 只有一匝和 (b) 10 匝时，环路的半径是多少？

Answer: (a) $a = 10$ cm, (b) $a = 1$ m. (See EM.)

练习 5-8：将一根导线形成一个方形环，并将其放置在 x-y 平面中，其中心位于原点，其每条边平行于 x 轴或 y 轴。每边长 40 厘米，导线承载 5 安培的电流，从上方看，电流方向为顺时针方向。计算环中心的磁场。

Answer: $\mathbf{H} = -\hat{\mathbf{z}}\dfrac{4I}{\sqrt{2\pi l}} = -\hat{\mathbf{z}}11.25$ A/m. (See EM.)



(a) Electric dipole (b) Magnetic dipole (c) Bar magnet

图 5-13 (a) 电偶极子的电场、(b) 磁偶极子的磁场和 (c) 条形磁铁的磁场的图案。远离源头，所有三种情况下的场模式都是相似的。